Reproducing the B2 rubidium-halide equations of state

Farla et al. (2025) | Numerical comparison and requests for original inputs

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Main finding. Direct EoSFit7c fits confirm the independent numerical implementation. Case 1 is close but does not reproduce every published coefficient digit. Constrained Case 3 agrees within 0.085 published marginal standard errors; small, rounding-compatible changes to fixed DAC anchors recover all four published thermal parameter pairs at printed precision.

Purpose and fitting setup

I am checking the implementation of the published equations of state in Peritheos. The method is described in the paper; the outstanding questions concern the exact numerical inputs, observation selection and saved fitting settings. Published coefficients and observations have been preserved separately from all diagnostic refits.

The model is third-order Birch-Murnaghan plus reference-subtracted Mie-Grüneisen-Debye thermal pressure, with $\gamma(V) = \gamma_0(V/V_0)^q$ and $\theta(V) = \theta_0 \exp[(\gamma_0 - \gamma(V))/q]$. The Debye energy reference is evaluated at the same volume and 300 K; n = 2 atoms per formula unit and Z = 1. Fits use GPa and cm³/mol.

The official EoSFit7c 7.60 executable (13 May 2021) was run directly. The principal selection contains all DAC observations and nonblank LVP observations above 300 K: 122 RbCl, 123 RbBr and 121 RbI points. This is an explicit reconstruction, not a recovery of the original use/omit flags.

1. Preferred Case 1

V_0 is fixed at 36.25, 41.74 and 51.11 cm³/mol. The comparison uses θ_0 = 177.04, 144.96 and 117.98 K from the public evaluation code; rounded θ_0 values were also tested. K_0, K_0', γ_0 and q are free.

Material / result	K_0 (GPa)	K_0'	γ_0	q
RbCl · published	19.89(8)	5.00(2)	1.96(4)	1.05(9)
RbCl · refit	19.92227	4.99511	1.96401	1.05929
RbBr · published	16.28(4)	5.28(2)	2.18(14)	1.52(24)
RbBr · refit	16.31590	5.25587	2.16658	1.47412
RbI · published	13.69(4)	4.95(1)	2.21(7)	1.42(10)
RbI · refit	13.75502	4.93392	2.19266	1.41792

Parentheses denote the published uncertainty in the last digits. The maximum absolute difference divided by the corresponding published marginal standard error is 0.40 for RbCl, 1.21 for RbBr and 1.63 for RbI. These are individual-parameter comparisons, not joint significance tests.

Fit uncertainties and constrained Case 3

Case 1: weighting and cross-checks

The weighted reconstruction uses reported P, V and T errors with effective variance $\sigma^2 = \sigma_P^2 + (\partial P/\partial V \cdot \sigma_V)^2 + (\partial P/\partial T \cdot \sigma_T)^2$. The independent calculation holds weights fixed during each inner solve and updates them between cycles. Printed-zero DAC pressure errors are retained; finite volume errors still produce finite effective weights. Original unrounded errors and any original correlation treatment remain unknown.

With matched rows, anchors, errors and R = 8.314 J mol^{-1} K^{-1}, native EoSFit and the independent refitter differ by at most 0.000326 GPa in K_0 and 0.000074 in the dimensionless parameters. Unweighted fits, inclusion of 300 K LVP rows, legacy volume conversion, and an illustrative 0.005 GPa pressure-error floor were also tested. None restores all Case 1 coefficient digits; the error floor materially changes the fit and is not adopted as a source uncertainty.

Native Case 1 standard errors	$\sigma(K_0)$, GPa	$\sigma(K_0')$	$\sigma(\gamma_0)$	$\sigma(q)$
RbCl	0.08601	0.02382	0.04225	0.08973
RbBr	0.04081	0.01514	0.13955	0.24376
RbI	0.03245	0.01107	0.06105	0.09112

These are EoSFit's rescaled errors for the diagnostic refits. Native covariance matrices and complete logs are retained; they do not replace the missing covariance of the published coefficients.

2. Case 3 with the literal printed anchors

Only γ_0 and q are refined. V_0, K_0, K_0' and θ_0 are fixed as follows; the two RbBr variants have different V_0 and K_0 as well as K_0'.

Case	V_0 (cm^3/mol)	K_0 (GPa)	K_0'	θ_0 (K)
RbCl	36.25	19.93	4.99	177
RbBr · DAC K_0'	41.74	16.32	5.26	145
RbBr · selected K_0'	41.55	17.55	4.98	145
RbI	51.11	13.68	4.96	118

Case	Published (γ_0, q)	Refit (γ_0, q)	Max. difference / source s.e.
RbCl	1.96(4), 1.04(9)	1.96029, 1.04704	0.078
RbBr · DAC K_0'	2.17(13), 1.49(23)	2.17459, 1.50152	0.050
RbBr · selected K_0'	2.11(18), 1.50(32)	2.11447, 1.50637	0.025
RbI	2.22(7), 1.43(10)	2.21407, 1.42931	0.085

Native refit standard errors ($\sigma\gamma_0$, σq): RbCl (0.04110, 0.08855); RbBr DAC (0.13941, 0.24233); RbBr selected K_0' (0.17655, 0.32106); RbI (0.06127, 0.09030). Fixed-anchor uncertainties are not propagated into this conditional two-parameter covariance.

Input precision and CsCl calibration

Case 3: diagnostic tests of unprinted anchor digits

The following native EoSFit examples change only one fitted DAC anchor within its printed rounding interval. All four thermal pairs then round to the publication. The deliberately selected RbBr K_0' = 4.98 is kept exact; that example varies V_0 instead.

Case	Illustrative anchor	Refit (γ_0, q)	Rounded pair
RbCl	K_0' = 4.989	1.95930, 1.04291	1.96, 1.04
RbBr · DAC K_0'	K_0' = 5.2575	2.17065, 1.48899	2.17, 1.49
RbBr · selected K_0'	V_0 = 41.549 cm³/mol	2.11426, 1.50362	2.11, 1.50
RbI	K_0' = 4.9606	2.21559, 1.43327	2.22, 1.43

Interpretation: these anchors were chosen after examining the discrepancies. They demonstrate compatibility with unprinted fixed-input precision; they are not estimates of the authors' actual values, unique solutions, or recovered author fits. This result does not establish Case 1 parity.

3. CsCl pressure calibration

The historical public EosCross v1.3 code supplies a Vinet static model plus integrated, reference-subtracted Debye thermal pressure. Inputs are K_0 = 16.74 GPa, K_0' = 5.703, θ_0 = 151 K, γ_0 = 1.99, q = 1.18, n = 2 and T_r = 300 K. The module V_0 is 42.179 cm³/mol, but the GUI overrides it from a_0 = 4.1221 Å using N_A = 6.0221367 × 10²³ mol^{-1}, giving V_0 = 42.1799615564 cm³/mol. These are recovered defaults, not a saved author session.

Execution of the pinned BurnMan 1.2.0 implementation agrees with the independent pressure calculation to 6.27 × 10^{-11} GPa across all 109 distinct CsCl readings. The same readings repeat in Tables S2-S4; they are not 327 independent calibration comparisons.

With printed T held fixed, 104 of 109 pressures agree within pressure and volume rounding: P ±0.005 GPa and V ±0.0005 Å³. The five remaining entries are:

Run	T (K)	V (Å³)	Printed P (GPa)	Replay P (GPa)
BT792	1100	60.468	5.54	5.53346
BT792	1200	60.901	5.49	5.51667
BT792	1300	61.389	5.44	5.48106
BT792	1400	61.773	5.44	5.51090
BT795	300	52.947	10.03	10.10028

Each printed pressure error is 0.07 GPa. All readings agree within that error plus the pressure/volume rounding allowance. The largest central-value difference is 0.07090 GPa. Temperature uncertainty and control tolerance are distinct from rounding; no inferred temperature correction is adopted. The batch code cubes the original lattice parameter, so the original unrounded lattice parameters, temperatures and saved calibration settings would help resolve these entries.

Calculated benchmark and requested files

4. RbCl Table S5: 5 GPa and 1500 K

Tables S5-S7 contain calculated V/V_0 benchmarks, not additional measured observations. I did not include these calculated values in the observational regressions. A numerical search finds one common coefficient vector within the printed half-digit bounds that reproduces all 54 ratios for RbBr, and another for RbI, within ±0.00005.

For RbCl, the search localizes the remaining discrepancy to 5 GPa and 1500 K. Omitting that entry allows all other 53 ratios to be reproduced by one coefficient vector that rounds to Table I. One diagnostic example is $(K_0, K_0', \gamma_0, q) = (19.88765988\ \text{GPa}, 5.00349408, 1.96362865, 1.04699136)$, with $V_0 = 36.25\ \text{cm}^3/\text{mol}$ and $\theta_0 = 177.04\ \text{K}$.

> At the omitted state, that reconstruction predicts **V/V_0 = 0.92828693**, rounding to **0.9283**; Table S5 prints **0.9282**. Direct evaluation of the rounded Table I coefficients instead gives about **0.9280678**, rounding to **0.9281**. The suggested 0.9283 therefore comes from joint reconstruction of the other 53 entries, not direct substitution of the printed coefficients.

Numerical searches subject to the other 53 rounding constraints give omitted-state ratios of approximately 0.92827583-0.92829062, all rounding to 0.9283. These are numerical search results, not certified global bounds. This is a request to check the entry against the original full-precision model, not a confirmed table correction. The published value remains preserved.

Files and details that would resolve the remaining questions

• **Original fitting files:** EoSFit data, initial/final models and refinement logs for Cases 1 and 3, including both RbBr Case 3 alternatives and the software version used.

• **Full-precision parameters:** the fixed DAC anchors, fixed Debye temperatures and fitted coefficients; parameter covariance matrices if available.

• **Selection and weighting:** observation use/omit flags, treatment of 300 K LVP data, original unrounded errors, enabled P/V/T uncertainty contributions and any correlations.

• **CsCl inputs:** original lattice parameters and temperatures, the exact calibrant file/reference lattice parameter, software version and any run-specific settings or corrections.

• **Benchmark calculation:** the full-precision model or calculation file used for Tables S5-S7, particularly the RbCl 5 GPa/1500 K entry.

The public supplement I located contains one DOCX; I did not identify an accessible original EoSFit/project archive in the public locations searched. A link would suffice if these files are deposited separately. Reconstructed inputs, native EoSFit models/logs and diagnostic covariance matrices are available from me for comparison.

Sources and scope

[1] Farla, Néri, Pöppelbaum & Glazyrin (2025), J. Appl. Phys. 137, 115901. doi:10.1063/5.0248905. Equation/parameter audit: DESY accepted manuscript; final publisher PDF was not accessed.
[2] Official AIP supplement, v1. doi:10.60893/figshare.jap.28515917.v1.
[3] Public EosCross code, commit 9518e36112be23504b84c4f595eab0b0207a3539 (23 August 2024). Historical source.
[4] Official EoSFit distribution; native refits performed 16 October 2026. This note assesses numerical reproducibility, not independent physical validation of extrapolated states.